

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA0501

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

Dr. A JAYA MABEL RANI

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	5
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	5
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	5
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	5
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	5
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	5

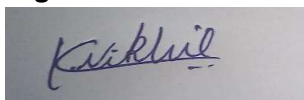
7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	5
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	5
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	5
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	5

Code of Conduct:

I K.Nikhil / 192424366 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

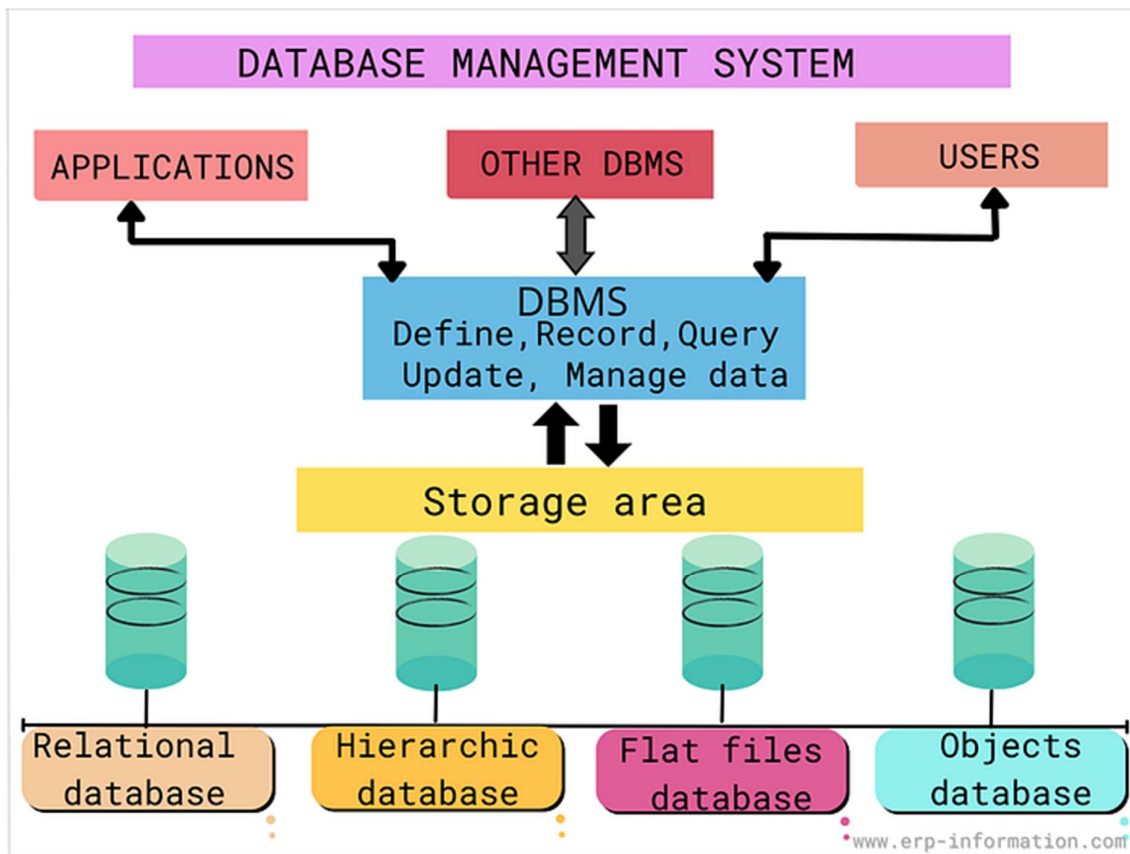
Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand



ANSWERS

NAME: K.NIKHIL LAKSHMI SAI

REG No. 192424366

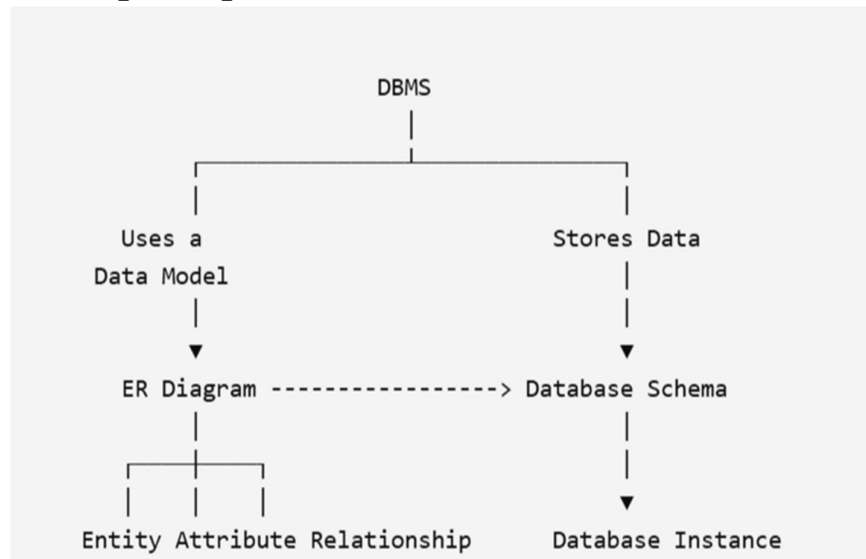
COURSE : CSA0501

1.A Database Management System (DBMS) is software used to store, organize, retrieve, and manage data efficiently. Before creating a database, a **Data Model** is selected to define how data is structured. The most commonly used model is the **Relational Data Model**.

An **ER (Entity-Relationship) Diagram** is created during database design. It represents real-world objects called **Entities**, their **Attributes**, and the **Relationships** between entities.

The ER diagram is converted into a **Database Schema**, which defines the logical structure of the database including tables, columns, and constraints. The actual data stored in the database at a particular moment is called an **Instance**.

Concept Map:



Term	Description
DBMS	Software used to manage databases
Data Model	Defines database structure
ER Diagram	Graphical representation of database design
Entity	Real-world object
Attribute	Property of an entity
Relationship	Association between entities
Schema	Logical design of database
Instance	Actual data stored at a specific time

Conclusion

The DBMS uses a data model to design an ER diagram, which is converted into a schema. The schema stores database instances containing actual data.

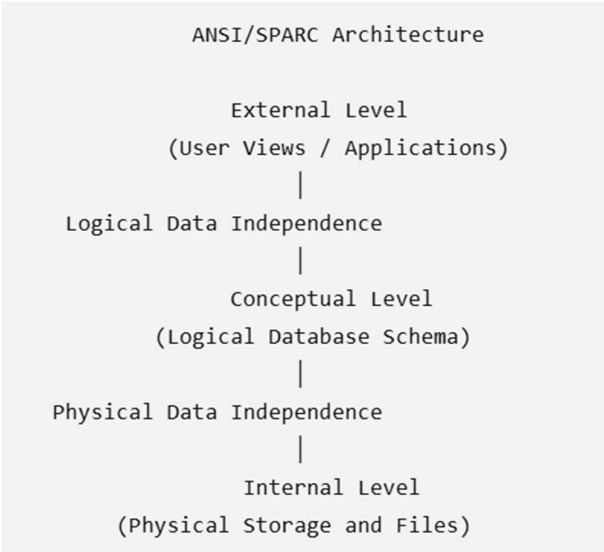
2.The ANSI/SPARC three-level architecture separates the database into three levels to provide **data independence**.

- 1. **External Level**
 - User views
 - Different users see different parts of the database.
- 2. **Conceptual Level**
 - Overall logical database structure.
 - Defines entities, relationships, and constraints.
- 3. **Internal Level**
 - Physical storage of data.
 - Includes files, indexes, and storage techniques.

Data Independence

- **Physical Data Independence**
 - Changes in the internal level do not affect the conceptual level.
- **Logical Data Independence**
 - Changes in the conceptual level do not affect external user views.

Concept Map



Table

Level	Description	Example
External	User-specific view	Student Portal
Conceptual	Overall logical database	Tables & Relationships
Internal	Physical storage	Disk files, Indexes

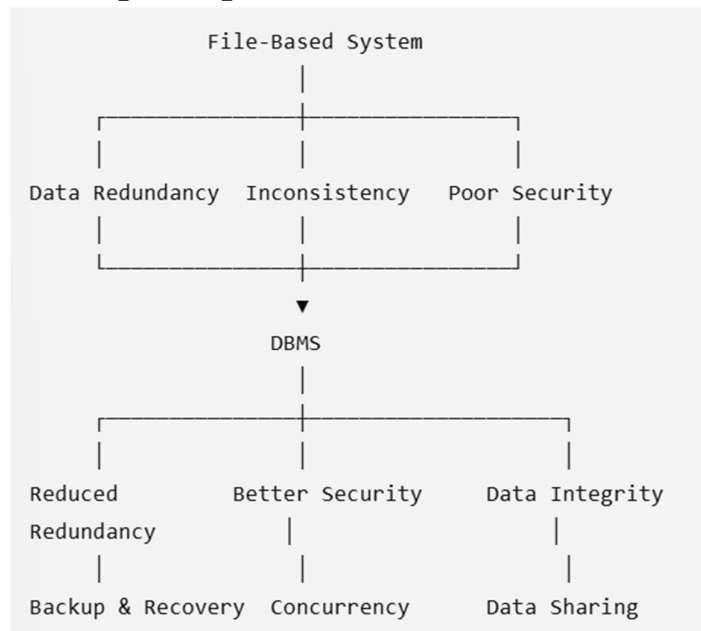
Conclusion

The ANSI/SPARC architecture improves security, flexibility, and database maintenance by separating user views, logical design, and physical storage.

3.A File-Based System stores data in separate files managed by individual applications. This often leads to duplicate data, inconsistency, and poor security.

A **Database Management System (DBMS)** centralizes data storage and provides efficient data management, security, concurrency control, and backup facilities.

Concept Map



Comparison Table

File-Based System	DBMS
Data redundancy	Reduced redundancy
Poor security	Strong security
Difficult sharing	Easy data sharing
No concurrency control	Supports concurrency
Difficult backup	Automatic backup and recovery
Data inconsistency	Improved consistency

Problems Solved by DBMS

- Reduces data redundancy
- Eliminates data inconsistency
- Improves security

- Enables data sharing
- Provides concurrency control
- Supports backup and recovery
- Maintains data integrity

Conclusion

A DBMS overcomes the limitations of file-based systems by providing secure, consistent, and efficient data management.

4. An Entity-Relationship (ER) Diagram is a graphical representation of a database. It helps identify entities, their attributes, relationships, and constraints before implementing the database.

ER Diagram Components

1. Entity

- A real-world object that can be identified uniquely.
- Example: Student, Employee, Product.

2. Weak Entity

- An entity that cannot be uniquely identified by its own attributes.
- Depends on a strong entity.
- Example: Dependent of an Employee.

3. Attribute Types

- **Simple Attribute:** Cannot be divided further (Age).
- **Composite Attribute:** Can be divided into sub-parts (Address → City, State, PIN).
- **Single-valued Attribute:** One value per entity (Roll Number).
- **Multi-valued Attribute:** Multiple values (Phone Numbers).
- **Derived Attribute:** Calculated from another attribute (Age from Date of Birth).

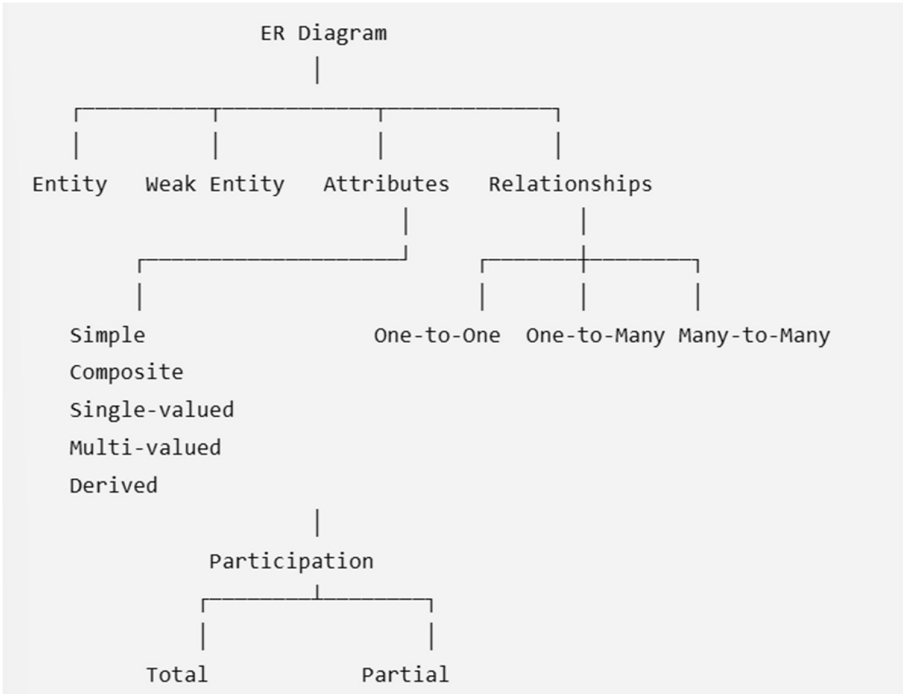
4. Relationship Types

- One-to-One (1:1)
- One-to-Many (1:M)
- Many-to-Many (M:N)

5. Participation Constraints

- **Total Participation:** Every entity must participate.
- **Partial Participation:** Participation is optional.

Concept Map



Table

Component	Description	Example
Entity	Independent object	Student
Weak Entity	Depends on another entity	Dependent
Simple Attribute	Cannot be divided	Age
Composite Attribute	Divisible	Address
Multi-valued Attribute	Multiple values	Phone Numbers
Derived Attribute	Computed	Age
Relationship	Connects entities	Student Enrolls Course
Participation	Mandatory/Optional	Total or Partial

Conclusion

ER diagrams provide a clear blueprint for database design by representing entities, relationships, attributes, and constraints.

5.A Data Model defines how data is organized, stored, and accessed in a database. The three major traditional data models are Hierarchical, Network, and Relational.

Hierarchical Model

- Tree structure
- Parent-child relationship
- One parent can have many children
- Supports one-to-many relationships

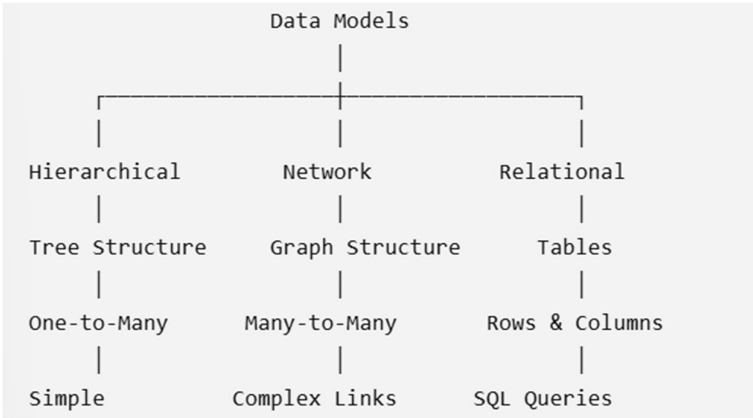
Network Model

- Graph structure
- Records connected through links
- Supports many-to-many relationships

Relational Model

- Data stored in tables (relations)
- Uses rows and columns
- Relationships established using Primary Key and Foreign Key
- Most widely used model

Concept Map



Comparison Table

Feature	Hierarchical	Network	Relational
Structure	Tree	Graph	Tables
Relationship	One-to-Many	Many-to-Many	Keys
Flexibility	Low	Medium	High
Complexity	Low	High	Medium
Query Language	Navigation	Navigation	SQL

Advantages

Hierarchical

- Simple structure
- Fast access

Network

- Flexible relationships
- Reduced redundancy

Relational

- Easy to understand
- Supports SQL
- High data independence
- Most commonly used

Conclusion

Among the three models, the **Relational Data Model** is the most popular because of its simplicity, flexibility, and SQL support.

6. The Enhanced Entity-Relationship (EER) Model extends the basic ER model by introducing advanced concepts such as **Generalization, Specialization, and Aggregation** to represent complex real-world scenarios.

EER Concepts

1. Generalization

- Combines similar lower-level entities into a higher-level entity.
- Example:
 - Car
 - Bike
 - Truck
 - Vehicle

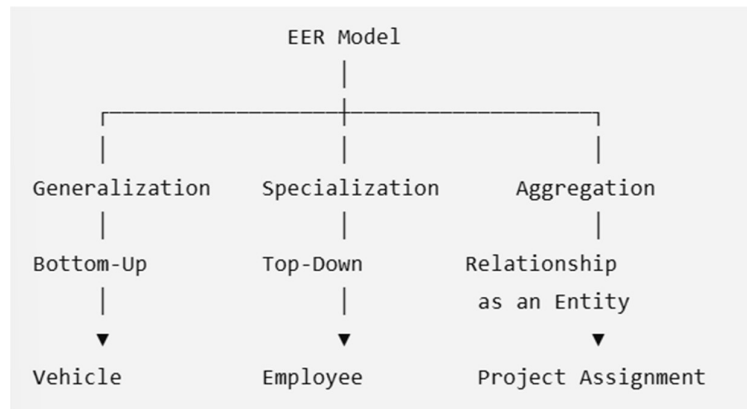
2. Specialization

- Divides one entity into multiple specialized entities.
- Example:
 - Employee
 - Manager
 - Engineer
 - Clerk

3. Aggregation

- Represents a relationship as a higher-level entity.
- Used when relationships participate in another relationship.

Concept Map



Table

Feature	Purpose	Example
Generalization	Combine entities	Vehicle
Specialization	Divide entity	Employee → Manager
Aggregation	Treat relationship as entity	Project Assignment

Difference Between ER and EER

ER Model	EER Model
Basic entities and relationships	Includes advanced concepts
Simpler design	More expressive design
Limited inheritance	Supports inheritance and abstraction

Conclusion

The EER model extends the ER model by supporting inheritance, abstraction, and complex relationships, making it suitable for large and real-world database applications.

7. The DBMS Software Architecture shows how a user's request is processed, optimized, executed, and finally retrieves data from the database.

Components

1. User

- Sends SQL queries through an application.

2. Application Program

- Accepts user input and forwards SQL commands to the DBMS.

3. Query Processor

- Parses SQL queries.
- Checks syntax and semantics.
- Optimizes the query for efficient execution.

4. Storage Manager

- Manages data files, indexes, and buffers.
- Retrieves the required data.

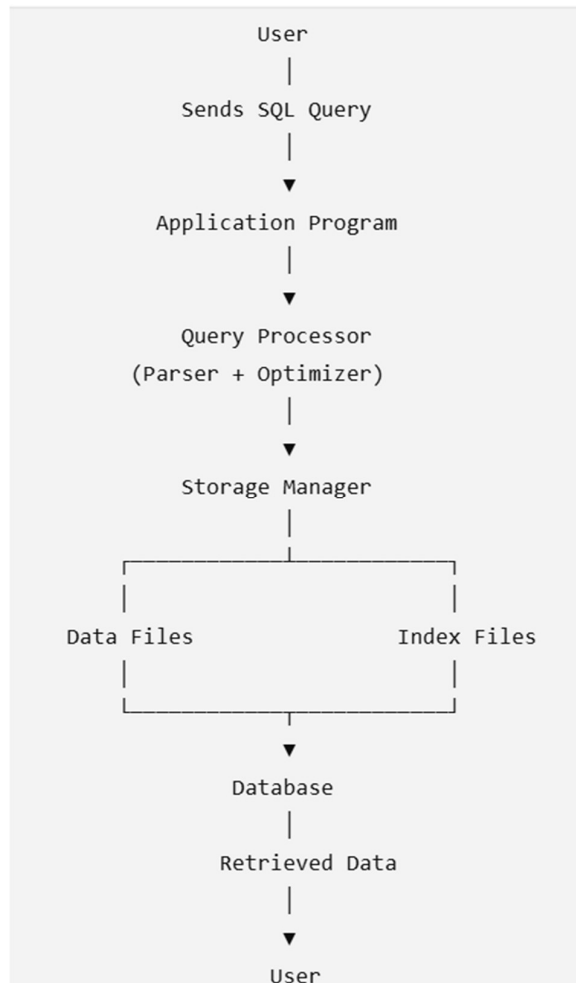
5. Database

- Stores the actual data permanently.

6. Result

- Retrieved data is returned to the user.

Concept Map



Table

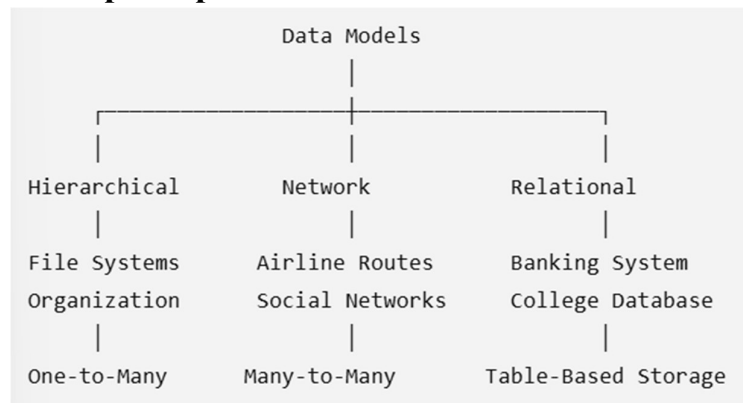
Component	Function
User	Sends request
Application	Accepts user input
Query Processor	Parses and optimizes SQL
Storage Manager	Retrieves required data
Database	Stores data

Conclusion

The DBMS architecture ensures efficient processing of user queries and fast retrieval of data.

8. Different data models are suitable for different real-world applications depending on how data is organized and related.

Concept Map



Applications Table

Data Model	Real-World Application	Reason for Suitability
Hierarchical	File System, Organization Chart	Tree structure with parent-child relationship
Network	Airline Route Network, Social Media	Supports many-to-many relationships
Relational	Banking, Hospital, College Management	Easy querying using SQL and reduced redundancy

Advantages

Hierarchical

- Fast access
- Simple navigation

Network

- Flexible connections
- Efficient for complex relationships

Relational

- Easy maintenance
- Supports SQL
- High security
- Data independence

Conclusion

Different data models are used based on application requirements, but the relational model is the most widely used due to its flexibility and SQL support.

9.A Schema defines the structure of a database, while an **Instance** represents the actual data stored at a particular time. Data independence ensures that changes at one level do not unnecessarily affect other levels.

Types of Schema

- **External Schema** – User-specific view.
- **Conceptual Schema** – Overall logical structure.
- **Internal Schema** – Physical storage details.

Data Independence

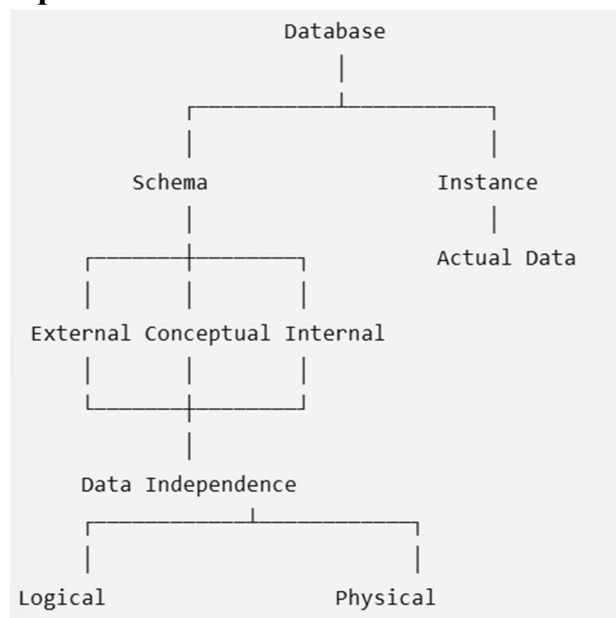
Logical Data Independence

- Changes in the conceptual schema do not affect external schemas.

Physical Data Independence

- Changes in the internal schema do not affect the conceptual schema.

Concept Map



Table

Concept	Description
Schema	Database design
Instance	Current database contents
External Schema	User view
Conceptual Schema	Logical database
Internal Schema	Physical storage
Logical Independence	Protects user views
Physical Independence	Protects logical structure

Conclusion

Schemas define the database structure, instances store actual data, and data independence improves flexibility and maintainability.

10.A DBMS environment includes different users who work together to design, manage, and use the database efficiently.

Roles

1. Database Administrator (DBA)

- Creates and manages databases.
- Controls security and user access.
- Performs backup and recovery.
- Monitors database performance.

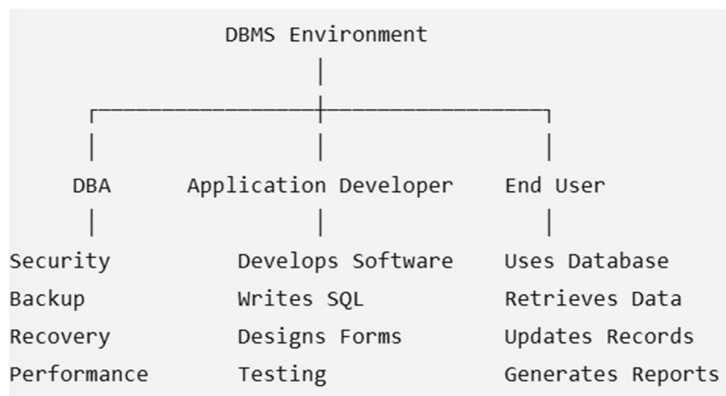
2. Application Developer

- Designs and develops database applications.
- Writes SQL queries and programs.
- Creates user interfaces.

3. End User

- Uses the application to insert, update, delete, and retrieve data.
- Does not manage the database directly.

Concept Map



Responsibilities Table

Role	Responsibilities
DBA	Security, backup, recovery, performance tuning
Application Developer	Develop applications, write SQL, testing
End User	Perform daily database operations

Conclusion

Each role has specific responsibilities, and together they ensure that the DBMS operates securely, efficiently, and meets user requirements.